

5 EXCEL SKILLS EVERY DATA ANALYST NEEDS



You do not need to learn everything in Excel first. Master these five skills, then build projects.

The fastest beginner roadmap for Excel in Data Analytics

Focus on lookup functions, PivotTables, data cleaning, formulas and charts. These are the practical Excel skills that show up repeatedly in analyst work, projects and interviews.

Inside this guide:

- 01 Lookup functions: XLOOKUP, VLOOKUP, INDEX-MATCH
- 02 PivotTables for quick summaries
- 03 Data cleaning basics
- 04 Core formulas and functions
- 05 Charts, visualisation and project ideas

Comment "EXCEL" to get roadmap, resources, cheat sheets and project ideas.

1. Lookup Functions + PivotTables

These two skills help you combine datasets and summarise large tables quickly. For data analysts, this is where Excel starts becoming a real analysis tool instead of just a spreadsheet.

Skill 1: Lookup functions

Function	Use it for
XLOOKUP	Search for an ID/name/code and return related values from another column.
VLOOKUP	Look up a value in the first column of a table and return a value from another column.
INDEX-MATCH	Build flexible lookups when the return column is not fixed or not to the right.

Example use case

Combine sales transactions with product category, region, manager or customer details using a common key.

Formula idea: =XLOOKUP(A2, ProductID, Category)

Practice project

Create an order table and a product master table, then use lookup functions to enrich the order table.

Skill 2: PivotTables

PivotTable task	Analyst benefit
Summarise by product	See top-selling categories and products.
Summarise by region	Compare performance by location.
Summarise by customer	Identify high-value customers.
Summarise by month	Spot trends over time.

2. Data Cleaning + Core Formulas

A lot of analyst work starts before analysis: fixing messy data. Once your dataset is clean, formulas help you create flags, segment rows, handle errors and build summary logic.

Skill 3: Data cleaning

Cleaning task	Why it matters
Remove duplicates	Prevents repeated rows from affecting counts and totals.
Handle blanks	Makes summaries and filters more reliable.
Clean text	Fixes spacing, casing and inconsistent names.
Fix dates	Allows proper monthly, weekly and trend analysis.
Standardise categories	Prevents region/product/customer categories from splitting incorrectly.

Safe cleaning rule

Before removing duplicates, keep a backup copy of the original data. Microsoft notes that Remove Duplicates permanently deletes duplicate data.

Skill 4: Formulas and functions

Formula	Use it for
IF	Create logic-based labels like Profit/Loss or Pass/Fail.
SUMIFS	Sum values based on one or more conditions.
COUNTIFS	Count rows that meet specific criteria.
IFERROR	Handle formula errors cleanly in dashboards.
Text/date functions	Clean names, split fields and analyse by month or year.

Formula practice

Build a sales summary using SUMIFS by region, product and month.

Formula idea: =SUMIFS(Sales, Region, G2)

Cleaning practice

Take a messy customer list and clean duplicates, blanks, text casing and dates.

3. Charts, Visualisation + Project Roadmap

Charts turn analysis into something stakeholders can understand. For beginner analysts, the goal is not fancy visuals. The goal is clarity.

Skill 5: Charts and visualisation

Visual	Use it for
Bar chart	Compare categories such as region, product or sales rep.
Line chart	Show trends over time.
Combo chart	Compare two related metrics on one view.
Conditional formatting	Highlight exceptions, top values or risk areas.
Dashboard layout	Combine KPIs, charts and slicers into a clean report.

4-week Excel project roadmap

Week	Focus	Output
Week 1	Lookup functions + clean dataset structure.	Enriched dataset using lookups.
Week 2	PivotTables + summary analysis.	Pivot report by product, region and month.
Week 3	Cleaning + formulas.	Cleaned file with formula-driven flags.
Week 4	Charts + dashboard storytelling.	One-page Excel dashboard + insights.

Project ideas

Sales dashboard, customer analysis, HR attrition, expense tracker, marketing campaign report, inventory summary.

Dashboard rule

Every dashboard should answer: What happened? Why did it happen? What should we do next?

Portfolio tip

Do not just upload a screenshot. Add a short write-up: business problem, data used, cleaning done, analysis method, key insights and recommendations.

4. Final Checklist + Free Resource Links

Use this final page as a quick checklist before moving to advanced Excel topics like Power Query, Power Pivot, advanced dashboards and automation.

Data Analyst Excel checklist

☐	Use XLOOKUP, VLOOKUP or INDEX-MATCH to combine datasets.
☐	Build PivotTables to summarise thousands of rows quickly.
☐	Clean duplicates, blanks, text, dates and inconsistent categories.
☐	Use IF, SUMIFS, COUNTIFS, IFERROR and text/date functions.
☐	Create clear charts and conditional formatting views.
☐	Build at least two Excel projects before moving to advanced topics.

Useful Microsoft links

VLOOKUP support	Microsoft VLOOKUP function
Remove duplicates	Find and remove duplicates
Unique/duplicate values	Filter for or remove duplicate values

Move next to advanced Excel

After these five skills, learn Power Query, Power Pivot, data models, dashboard design and automation.

Best practice

Every Excel project should include raw data, cleaned data, analysis, charts and written insights.

Comment "EXCEL"

and I will send you a complete Excel roadmap, practice resources, cheat sheets and project ideas for Data Analytics.
Follow [@analyticswith_sundeeep](#) for practical data analytics career resources.